

Exercise 2: Implementing the Factory Method Pattern

Scenario:

You are developing a document management system that needs to create different types of documents (e.g., Word, PDF, Excel). Use the Factory Method Pattern to achieve this.

```
using System;
```

```
interface IDocument
```

```
{  
    void Display();  
}
```

```
class WordDocument : IDocument
```

```
{  
    public void Display()  
    {  
        Console.WriteLine("This is a Word Document");  
    }  
}
```

```
class PdfDocument : IDocument
```

```
{  
    public void Display()  
    {  
        Console.WriteLine("This is a PDF Document");  
    }  
}
```

```
class ExcelDocument : IDocument
```

```
{  
    public void Display()
```

```
{  
    Console.WriteLine("This is an Excel Document");  
}  
}
```

abstract class DocumentFactory

```
{  
    public abstract IDocument CreateDocument();  
}
```

// Concrete Factories

class WordFactory : DocumentFactory

```
{  
    public override IDocument CreateDocument()  
    {  
        return new WordDocument();  
    }  
}
```

class PdfFactory : DocumentFactory

```
{  
    public override IDocument CreateDocument()  
    {  
        return new PdfDocument();  
    }  
}
```

class ExcelFactory : DocumentFactory

```
{  
    public override IDocument CreateDocument()  
    {
```

```
        return new ExcelDocument();
    }
}

// Client
class FactoryMethodPatternExample
{
    static void Main(string[] args)
    {
        DocumentFactory factory;

        factory = new WordFactory();
        IDocument doc1 = factory.CreateDocument();
        doc1.Display();

        factory = new PdfFactory();
        IDocument doc2 = factory.CreateDocument();
        doc2.Display();

        factory = new ExcelFactory();
        IDocument doc3 = factory.CreateDocument();
        doc3.Display();
    }
}
```

```
Microsoft Visual Studio Debug Console
+ v

This is a Word Document
This is a PDF Document
This is an Excel Document

C:\Users\HP\source\repos\FactoryMethodPatternExample\FactoryMethodPatternExample\bin\Debug\net10.0\FactoryMethodPatternE
xample.exe (process 16864) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```